

Nonlinear Longitudinal Flight Dynamics Modeling and Trim Analysis of a Cessna 182

A 3-Degree-of-Freedom Simulation Study

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Abstract

This report presents the development, verification, and analysis of a nonlinear, three-degree-of-freedom (3-DOF) longitudinal flight dynamics model for a Cessna 182 aircraft. Starting from the general 6-DOF flat-Earth rigid-body equations of motion, the model is reduced under a longitudinal-symmetry assumption to six states. A distinguishing feature of this implementation is the treatment of the unsteady aerodynamic term $\dot{\alpha}$ (angle-of-attack rate): rather than neglecting its contribution to lift and pitching moment, or approximating it with a delayed/previous-step estimate, the coupled algebraic relationship among $\dot{u}$, $\dot{w}$, and $\dot{\alpha}$ is solved in closed form as a 3×3 linear system at every time step. The model is implemented in MATLAB and integrated into a Simulink simulation environment. Straight-and-level trim conditions are computed at 5,000 ft for true airspeeds of 30–80 m/s using a penalty-based unconstrained numerical optimization. Starting from the 60 m/s trim point, an elevator doublet maneuver is simulated and the resulting time histories are presented, exhibiting the expected lightly-damped phugoid response characteristic of a statically stable aircraft.

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1 Introduction

The objective of this project is to develop a nonlinear longitudinal (3-DOF) flight dynamics simulation of a Cessna 182 aircraft, using the aerodynamic, inertial, and geometric data summarized in Table 1. The work is organized to address, in order: (i) the architecture of the simulation and its sub-components, (ii) the governing equations of motion and their derivation, (iii) the assumptions and literature-based data used to fill gaps in the given dataset, (iv) a walkthrough of the implemented code, (v) a simple visualization of the aircraft's motion, (vi) a trim (equilibrium) analysis across a range of airspeeds, and (vii) a closed-loop-free dynamic response to an elevator doublet input, starting from a computed trim condition.

Table 1: Cessna 182 nominal aircraft data used throughout this study.

Quantity	Symbol	Value
Wing reference area	S_{ref}	16.2 m ²
Mean aerodynamic chord	$\bar{c}$	1.49 m
Wing span	b	11 m
Mass	m	1202 kg
Moment of inertia (roll)	I_{xx}	1285 kg m ²
Moment of inertia (pitch)	I_{yy}	1825 kg m ²
Moment of inertia (yaw)	I_{zz}	2667 kg m ²
Product of inertia	I_{xz}	0 kg m ²
C_{D_0}		0.0270
C_{D_α}		0.121
$C_{D_{\delta_e}}$		0
C_{L_0}		0.307
C_{L_α}		4.41
$C_{L_{\dot{\alpha}}}$		1.7
C_{L_q}		3.9
$C_{L_{\delta_e}}$		0.43
C_{m_0}		0.04
C_{m_α}		-0.613
$C_{m_{\dot{\alpha}}}$		-7.27
C_{m_q}		-12.4
$C_{m_{\delta_e}}$		-1.122

2 Software and Libraries

The simulation is implemented entirely in MATLAB/Simulink. The following built-in capabilities are used:

- `ode45` (variable-step, 4th/5th-order Runge–Kutta) for time integration of the nonlinear equations of motion, both in stand-alone MATLAB scripts and inside Simulink (via the model's solver configuration).
- `fminsearch` (derivative-free Nelder–Mead simplex) for the trim/equilibrium search described in Section 9.
- The backslash operator (`Mmat\rhs`) for the closed-form linear solve of the coupled $\dot{\mathbf{x}}$ system described in Section 6.6.
- Simulink blocks (`Step`, `Sum`, `Rate Limiter`, `Interpreted MATLAB Function`, `Integrator`, `To Workspace`, `Scope`) for the closed-form dynamic simulation and doublet maneuver of

Section 10.

No third-party toolboxes are required; all functionality is standard MATLAB/Simulink.

3 Coordinate Systems and Assumptions

The vehicle-body axis system (x_b forward, y_b right, z_b down) is used throughout, following the standard aircraft flight-dynamics convention (Stevens & Lewis, 2015). The following assumptions were made in reducing the general 6-DOF rigid-body equations of motion to the 3-DOF longitudinal model implemented here.

3.1 Structural / Kinematic Assumptions

- A1. Flat, non-rotating Earth.** Earth's rotation rate $\omega_{i/e}$ and curvature are neglected. The vehicle-carried NED frame F_v is treated as inertial ($F_v \equiv F_i$), which is valid because the flight distances and durations considered here are small relative to Earth's radius.
- A2. Rigid body.** The aircraft is treated as a single rigid body; no structural flexibility or aeroelastic effects are modeled.
- A3. Constant mass.** Fuel consumption is neglected over the simulation timespan; m is treated as constant (no $\dot{m}$ term in Newton's second law).
- A4. Longitudinal symmetry.** The aircraft remains in its x - z symmetry plane at all times: $v = 0$, $p = r = 0$, $\phi = \psi = 0$, $\beta = 0$, $P_E = 0$. This reduces the general 12-state, 6-DOF model to a 6-state, 3-DOF longitudinal model.
- A5. $I_{xz} = 0$.** Given directly in the aircraft data table; combined with A4, this guarantees zero cross-coupling between the longitudinal and lateral-directional dynamics.

3.2 Aerodynamic Model Assumptions

- A6. Linearized aerodynamic model.** C_L , C_D , C_m are modeled as linear (first-order Taylor) functions of α , $\dot{\alpha}$, $\dot{q}$, and δ_e , valid for small perturbations about a reference condition and pre-stall angles of attack.
- A7. No elevator-induced drag.** $C_{D_{\delta_e}} = 0$ is given directly in the data table.
- A8. Explicit (non-neglected) unsteady $\dot{\alpha}$ coupling.** Unlike a simplified treatment that discards the effect of $\dot{\alpha}$ on lift to force an explicit ODE, this model retains $C_{L_{\dot{\alpha}}}$ and $C_{m_{\dot{\alpha}}}$ fully and resolves the resulting implicit relationship as a closed-form linear system, solved at every evaluation of the state derivative (Section 6.6).
- A9. Constant stability derivatives.** All aerodynamic derivatives are treated as constants over the flight envelope considered, independent of α , V , or Mach number.
- A10. Incompressible flow.** No Mach-number correction is applied, consistent with the low-speed operating regime of this aircraft class.
- A11. No propeller slipstream effect on the airframe.** The propulsive influence on the aircraft is captured only as a direct thrust force (A14); any indirect effect of the slipstream on wing/tail lift, drag, or pitching moment is not modeled, due to lack of available data.

3.3 Propulsion Model Assumptions

- A12. Thrust acts purely along the body x -axis, through the CG.** No vertical or lateral thrust-line offset is modeled, i.e., $\bar{F}_E^b = [T, 0, 0]^T$ and thrust contributes zero moment, since no published offset data exists for this airframe.
- A13. Constant propeller efficiency.** A single representative value $\eta_p = 0.80$ is used across all airspeeds and throttle settings, rather than a full advance-ratio-dependent propeller map (see Section 6.3).

3.4 Force/Moment Assembly and Position-State Assumptions

- A14. Additive force decomposition.** $\bar{F}^b = \bar{F}_g^b + \bar{F}_E^b + \bar{F}_A^b$: gravity, engine, and aerodynamic contributions are summed independently in the body frame, each rotated in from its natural reference frame.
- A15. Moments taken about the CG.** Since all moments are referenced to the center of gravity, $\bar{M}_g^b \equiv 0$ identically; combined with A14, the total moment reduces to the aerodynamic pitching moment alone.
- A16. Lateral position P_E not required.** Consistent with A4, lateral position never enters the dynamics and is omitted from the state vector.
- A17. Altitude-dependent dynamics.** Unlike the general rigid-body case (which is position-invariant), P_D is retained as a state *and* feeds back into the dynamics, since air density $\rho(h)$ (Section 6.1) and, in this implementation, gravitational acceleration $g(h)$ (Section 6.2) both depend on altitude $h = -P_D$.
- A18. ψ equation not required.** The model performs identically regardless of heading; only ϕ and θ affect the dynamics through the gravity-rotation term, and ϕ is already fixed at zero by A4.

4 System Architecture

The simulation is organized as a chain of physically distinct sub-systems, each responsible for a single stage of the force/moment build-up, connected through explicit state-feedback taps rather than global variables. This data-flow representation follows the extended design structure matrix (XDSM) convention of Lambe & Martins (2012), in which process blocks are laid out along a diagonal-like flow and both forward data dependencies and feedback loops are shown explicitly.

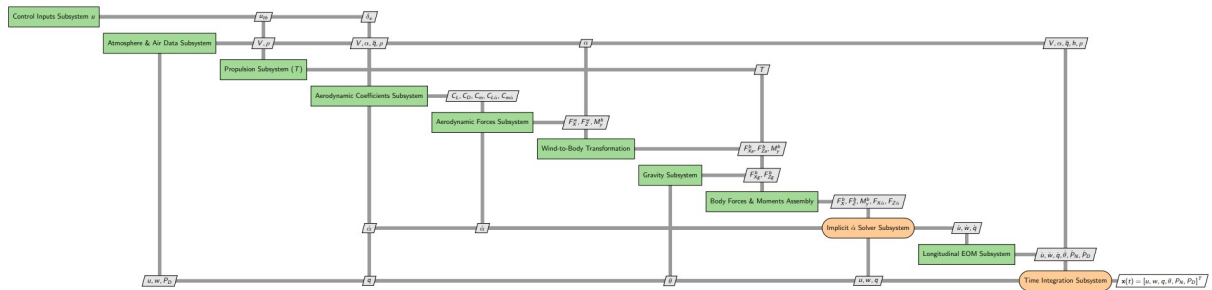


Figure 1: System architecture of the nonlinear Cessna 182 longitudinal simulation. Green blocks denote purely algebraic (feed-forward) sub-systems; orange blocks denote the implicit/coupled solve and the time-integration stage.

The nine sub-systems, in data-flow order, are:

1. **Control Inputs** – supplies δ_e (elevator, rad) and u_{th} (throttle, $[0, 1]$).
2. **Atmosphere & Air Data** – computes V , α , $\bar{q}$, ρ , and h from the current state.
3. **Propulsion** – computes thrust T from V and u_{th} .
4. **Aerodynamic Coefficients** – computes C_L , C_D , C_m (split into $\dot{\alpha}$ -independent and $\dot{\alpha}$ -dependent parts).
5. **Aerodynamic Forces** – converts coefficients to dimensional lift, drag, and pitching moment in wind axes.
6. **Wind-to-Body Transformation** – rotates aerodynamic forces into the body frame via α .
7. **Gravity** – rotates the weight vector into the body frame via θ .
8. **Body Forces & Moments Assembly** – sums all force/moment contributions.
9. **Implicit $\dot{\alpha}$ Solver and Longitudinal EOM** – jointly resolve $\dot{u}$, $\dot{w}$, $\dot{\alpha}$, $\dot{q}$, and integrate the full state (Sections 5–6.6).

State-feedback paths (bottom of Fig. 1) explicitly show which sub-system consumes which state component – e.g., the Aerodynamic Coefficients block requires u, w, q (to form α , V , and $\hat{q}$), while the Longitudinal EOM block additionally requires θ (for the position kinematics).

5 Equations of Motion

5.1 General Rigid-Body Flat-Earth Equations

The full derivation begins from Newton's second law for translation and rotation, applied in the body frame via the Coriolis theorem, following the standard flat-Earth formulation (Stevens & Lewis, 2015; Lum, AE512 Lecture 04b).

Translational velocity. Newton's second law in the inertial frame,

$$\bar{F} = \frac{d}{dt} \Big|_i [m\bar{V}] = m^i \dot{\bar{V}}, \quad (1)$$

is transformed into the body frame using the Coriolis relation ${}^i \dot{\bar{V}} = {}^b \dot{\bar{V}} + \bar{\omega}_{b/i} \times \bar{V}$, giving

$${}^b \dot{\bar{V}}^b = \begin{pmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \end{pmatrix} = \frac{1}{m} \bar{F}^b - \bar{\omega}_{b/i}^b \times \bar{V}^b. \quad (2)$$

Rotational velocity. Applying Newton's second law for rotation, $\bar{M} = \frac{d}{dt} \Big|_i [\bar{H}]$ with $\bar{H} = I_b \bar{\omega}_{b/i}$, and again invoking the Coriolis theorem (this time because I_b is constant only when viewed from the body frame),

$${}^b \dot{\bar{\omega}}_{b/i}^b = I_b^{-1} \left[\bar{M}^b - \bar{\omega}_{b/i}^b \times I_b \bar{\omega}_{b/i}^b \right]. \quad (3)$$

Angular position. From the Euler kinematical equations,

$$\dot{\Phi} = H(\Phi) \bar{\omega}_{b/v}^b, \quad H(\Phi) = \begin{pmatrix} 1 & \tan \theta \sin \phi & \tan \theta \cos \phi \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi / \cos \theta & \cos \phi / \cos \theta \end{pmatrix}. \quad (4)$$

Translational position. From the navigation equations,

$$\dot{\bar{P}}_{NED} = C^{vb}(\Phi) \bar{V}^b. \quad (5)$$

Equations (2)–(5) together form the general 12-state, 6-DOF flat-Earth equations of motion, valid for any rigid body – the only vehicle-specific terms are m , I_b , $\bar{F}^b$, and $\bar{M}^b$.

5.2 Reduction to 3-DOF Longitudinal Form

Applying assumptions A1–A5 ($v = 0$, $p = r = 0$, $\phi = \psi = 0$, $P_E = 0$, $I_{xz} = 0$), Eq. (2) reduces to

$$\dot{u} = \frac{F_x}{m} - qw, \quad \dot{w} = \frac{F_z}{m} + qu, \quad (6)$$

Eq. (3) reduces to a single scalar equation (all gyroscopic cross-coupling vanishes identically for $p = r = 0$, $I_{xz} = 0$):

$$\dot{q} = \frac{M_y}{I_{yy}}, \quad (7)$$

Eq. (4) reduces to $\dot{\theta} = q$, and Eq. (5) reduces to

$$\dot{P}_N = u \cos \theta + w \sin \theta, \quad \dot{P}_D = -u \sin \theta + w \cos \theta. \quad (8)$$

The resulting 3-DOF longitudinal state-space model is

$$\dot{\bar{x}} = \begin{pmatrix} \dot{u} \\ \dot{w} \\ \dot{q} \\ \dot{\theta} \\ \dot{P}_N \\ \dot{P}_D \end{pmatrix} = \bar{f}(\bar{x}, \bar{u}), \quad \bar{x} = \begin{pmatrix} u \\ w \\ q \\ \theta \\ P_N \\ P_D \end{pmatrix}, \quad \bar{u} = \begin{pmatrix} \delta_e \\ u_{th} \end{pmatrix}. \quad (9)$$

The remaining task – and the primary technical content of this project – is to express $\bar{F}^b = [F_x, 0, F_z]^T$ and M_y in terms of $\bar{x}$, $\bar{u}$, and constants.

6 Sub-System Models

6.1 Atmosphere Model (ISA)

Air density is computed via the standard International Standard Atmosphere (ISA) troposphere model, valid up to 11,000 m:

$$T(h) = T_0 - Lh, \quad \rho(h) = \rho_0 \left(\frac{T(h)}{T_0} \right)^{\frac{g_0}{RL} - 1}, \quad (10)$$

with $\rho_0 = 1.225 \text{ kg/m}^3$, $T_0 = 288.15 \text{ K}$, $L = 0.0065 \text{ K/m}$, $R = 287.05 \text{ J/(kg K)}$, $g_0 = 9.80665 \text{ m/s}^2$. This is more than sufficient for the Cessna 182's operational ceiling ($\approx 5,500 \text{ m}$).

6.2 Gravity Model

Rather than treating g as a constant, gravitational acceleration is modeled as decreasing with altitude following the inverse-square law:

$$g(h) = g_0 \left(\frac{R_e}{R_e + h} \right)^2, \quad (11)$$

with $R_e = 6,371,000 \text{ m}$ (mean Earth radius).

6.3 Propulsion Model

The Cessna 182 (as certified) is equipped with a Lycoming IO-540-AB1A5 engine rated at 230 hp (EASA Type Certificate Data Sheet EASA.IM.A.052), converted to SI units:

$$P_{\max} = 230 \times 745.7 \text{ W} \approx 171.5 \text{ kW}. \quad (12)$$

Following a constant-efficiency propeller model (consistent with, e.g., Virginia Tech AOE 3104 course notes on propeller thrust models), the propulsive power is $P_{\text{prop}} = \eta_p P_{\text{shaft}}$ with $P_{\text{shaft}} = u_{\text{th}} P_{\max}$, and since $P_{\text{prop}} = TV$,

$$T = \frac{\eta_p u_{\text{th}} P_{\max}}{V}. \quad (13)$$

A propeller efficiency of $\eta_p = 0.80$ is assumed (representative literature value for a fixed-pitch general-aviation propeller near cruise conditions; no published efficiency map exists for this specific airframe/engine combination). Thrust acts purely along the body x -axis (A14): $\bar{F}_E^b = [T, 0, 0]^T$.

6.4 Aerodynamic Coefficients and Loads

Following the standard linear stability-derivative model,

$$C_L = C_{L_0} + C_{L_\alpha} \alpha + C_{L_{\hat{\alpha}}} \hat{\alpha} + C_{L_{\hat{q}}} \hat{q} + C_{L_{\delta_e}} \delta_e, \quad (14)$$

$$C_D = C_{D_0} + C_{D_\alpha} \alpha, \quad (15)$$

$$C_m = C_{m_0} + C_{m_\alpha} \alpha + C_{m_{\hat{\alpha}}} \hat{\alpha} + C_{m_{\hat{q}}} \hat{q} + C_{m_{\delta_e}} \delta_e, \quad (16)$$

where $\hat{\alpha} = \dot{\alpha} \bar{c} / (2V)$ and $\hat{q} = q \bar{c} / (2V)$ are the standard non-dimensional rate terms. Dimensional loads follow from

$$L = \bar{q} S C_L, \quad D = \bar{q} S C_D, \quad M_y = \bar{q} S \bar{c} C_m, \quad \bar{q} = \frac{1}{2} \rho V^2. \quad (17)$$

6.5 Coordinate Transformations and Force Assembly

The aerodynamic force, expressed in wind axes as $\bar{F}_A^w = [-D, 0, -L]^T$, is rotated into the body frame using the wind-to-body rotation (a function of α only, since $\beta = 0$):

$$C^{wb}(\alpha) = \begin{pmatrix} \cos \alpha & 0 & -\sin \alpha \\ 0 & 1 & 0 \\ \sin \alpha & 0 & \cos \alpha \end{pmatrix}, \quad \bar{F}_A^b = C^{wb}(\alpha) \bar{F}_A^w. \quad (18)$$

Gravity is rotated from the vertical (NED) frame using $\bar{F}_g^v = [0, 0, mg]^T$ and the (simplified, $\phi = \psi = 0$) rotation

$$C^{bv}(\theta) = \begin{pmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{pmatrix}, \quad \bar{F}_g^b = C^{bv}(\theta) \bar{F}_g^v. \quad (19)$$

The total body-axis force and moment are then

$$\bar{F}^b = \bar{F}_A^b + \bar{F}_E^b + \bar{F}_g^b, \quad M_y = M_{y,A} \quad (\text{since } M_{y,g} = 0, M_{y,E} = 0 \text{ by A12, A14}). \quad (20)$$

6.6 The Coupled $\dot{\alpha}$ Problem and its Closed-Form Solution

6.6.1 The algebraic loop

Because C_L and C_m depend on $\dot{\alpha}$, and $\dot{\alpha}$ itself is defined kinematically from $\dot{u}$ and $\dot{w}$,

$$\dot{\alpha} = \frac{u\dot{w} - w\dot{u}}{V^2}, \quad (21)$$

while $\dot{u}$ and $\dot{w}$ in turn depend on C_L (and hence on $\dot{\alpha}$) through the lift force, the system is *implicit*: $0 = F(\dot{x}, \bar{x}, \bar{u})$ rather than the explicit form $\dot{x} = f(\bar{x}, \bar{u})$ required by standard ODE solvers such as `ode45`. A common simplification is to neglect the $\dot{\alpha}$ contribution to L (setting $\hat{\alpha} = 0$), which restores an explicit form but discards a genuine unsteady-aerodynamic effect. In this work, the implicit system is instead solved exactly, in closed form, at every function evaluation.

6.6.2 Closed-form derivation

C_L and C_m are first split into an $\dot{\alpha}$ -independent (“base”) part and a term linear in $\dot{\alpha}$:

$$C_L = \underbrace{C_{L_0} + C_{L_\alpha} \alpha + C_{L_q} \hat{q} + C_{L_{\delta_e}} \delta_e}_{C_{L,\text{base}}} + C_{L_{\dot{\alpha}}} \frac{\bar{c}}{2V} \dot{\alpha}, \quad (22)$$

and similarly for C_m . This propagates linearly through the lift force and, after rotation into the body frame, through F_x and F_z (drag has no $\dot{\alpha}$ -dependence, but the rotated lift term contributes to *both* F_x and F_z via $\sin \alpha$ and $\cos \alpha$):

$$F_x = F_{x,\text{base}} + F_{x,\text{coef}} \dot{\alpha}, \quad F_z = F_{z,\text{base}} + F_{z,\text{coef}} \dot{\alpha}, \quad (23)$$

with $F_{x,\text{coef}}$ and $F_{z,\text{coef}}$ obtained from $L_{\text{coef}} = \bar{q} S C_{L_{\dot{\alpha}}} \bar{c} / (2V)$ through the same wind-to-body rotation. Substituting into Eq. (2) gives $\dot{u}$ and $\dot{w}$ as affine functions of $\dot{\alpha}$:

$$\dot{u} = A_1 + B_1 \dot{\alpha}, \quad A_1 = \frac{F_{x,\text{base}}}{m} - q w, \quad B_1 = \frac{F_{x,\text{coef}}}{m}, \quad (24)$$

$$\dot{w} = A_2 + B_2 \dot{\alpha}, \quad A_2 = \frac{F_{z,\text{base}}}{m} + q u, \quad B_2 = \frac{F_{z,\text{coef}}}{m}. \quad (25)$$

Substituting these into the definition of $\dot{\alpha}$ itself (Eq. (21)), multiplied through by V^2 ,

$$u\dot{w} - w\dot{u} - V^2 \dot{\alpha} = 0, \quad (26)$$

yields three linear equations in the three unknowns $\dot{u}$, $\dot{w}$, $\dot{\alpha}$:

$$\underbrace{\begin{pmatrix} 1 & 0 & -B_1 \\ 0 & 1 & -B_2 \\ -w & u & -V^2 \end{pmatrix}}_M \begin{pmatrix} \dot{u} \\ \dot{w} \\ \dot{\alpha} \end{pmatrix} = \begin{pmatrix} A_1 \\ A_2 \\ 0 \end{pmatrix}. \quad (27)$$

This is solved directly (`sol = M\rhs` in MATLAB) at every evaluation of the state derivative – no iteration, no history dependence, and no approximation of the $\dot{\alpha}$ term is required. Once $\dot{\alpha}$ is known, the pitching moment is completed as $M_y = M_{y,\text{base}} + M_{\text{coef}} \dot{\alpha}$ (with $M_{\text{coef}} = \bar{q} S C_{m_{\dot{\alpha}}} \bar{c} / (2V)$), and $\dot{q} = M_y / I_{yy}$ follows directly.

6.6.3 Final Closed-Form State Derivative

Collecting the results of this section together with Eqs. (2)–(5) as reduced in Section 5, the complete 3-DOF longitudinal state derivative implemented in `cessna182_model.m` is

$$\dot{\bar{x}} = \begin{pmatrix} \dot{u} \\ \dot{w} \\ \dot{q} \\ \dot{\theta} \\ \dot{P}_N \\ \dot{P}_D \end{pmatrix} = \begin{pmatrix} \frac{F_x^b}{m} - qw \\ \frac{F_z^b}{m} + qu \\ \frac{M_y^b}{I_{yy}} \\ q \\ u \cos \theta + w \sin \theta \\ -u \sin \theta + w \cos \theta \end{pmatrix}, \quad (28)$$

where F_x^b , F_z^b , and M_y^b are now understood to already include the closed-form $\dot{\alpha}$ contribution, obtained by solving Eq. (27) for $\dot{u}$, $\dot{w}$, $\dot{\alpha}$ and completing $M_y^b = M_{y,\text{base}} + M_{\text{coef}}\dot{\alpha}$. This is the exact function $\bar{f}(\bar{x}, \bar{u})$ evaluated by the solver at every time step, with no approximation beyond the linearized aerodynamic model of Section 3.

7 Code Walkthrough

The complete state-derivative function is listed below. Its structure follows Section 4 directly: air data, altitude-dependent atmosphere and gravity, aerodynamic-coefficient splitting, propulsion, coordinate transformations, and the closed-form $\dot{\alpha}$ solve, in that order.

Listing 1: `cessna182_model.m` – nonlinear longitudinal state-derivative function.

```

1 function xdot = cessna182_model(x,u)
2
3 % CESSNA182_MODEL - Longitudinal 3DOF Equations of Motion
4 %
5 % States:   x = [u; w; q; theta; PN; PD]
6 % Controls: u = [de; uth]   (elevator deflection [rad], throttle [0-1])
7
8 %----- NOMINAL VEHICLE CONSTANTS -----
9 m      = 1202;           % Aircraft mass (kg)
10 cbar   = 1.49;           % Mean aerodynamic chord (m)
11 S      = 16.2;          % Wing planform area (m^2)
12 b      = 11;            % Wing span (m)
13
14 Iyy    = 1825;           % kg*m^2
15
16 g0     = 9.80665;        % m/s^2, sea-level standard gravity
17 Re     = 6371000;        % m, mean Earth radius (for g(h) inverse-square
    law)
18
19 % Aerodynamic derivatives
20 CD0    = 0.0270;
21 CDa    = 0.121;
22 CDde   = 0;
23 CL0    = 0.307;
24 CLa    = 4.41;
25 CLadot = 1.7;
26 CLq    = 3.9;
27 CLde   = 0.43;
28 Cm0    = 0.04;

```

```

29 Cma      = -0.613;
30 Cmadot   = -7.27;
31 Cmq      = -12.4;
32 Cmde     = -1.122;
33
34 % Propulsion (Lycoming IO-540-AB1A5, EASA TCDS EASA.IM.A.052)
35 Pmax     = 230*745.7;           % Max shaft power, converted to SI (W)
36 etap     = 0.80;               % Propeller efficiency (assumed, literature)
37
38 %-----STATE AND CONTROL VECTORS
39
40 -----
41 uvel = x(1);           % body-axis x velocity
42 wvel = x(2);           % body-axis z velocity
43 q      = x(3);           % pitch rate
44 theta  = x(4);           % pitch angle
45 PN     = x(5);           % north position
46 PD     = x(6);           % down position (negative altitude)
47
48 de      = u(1);           % elevator deflection (rad)
49 uth     = u(2);           % throttle setting [0,1]
50
51 %-----AIR DATA CALCULATIONS
52
53 -----
54 V      = sqrt(uvel^2 + wvel^2);
55 alpha  = atan2(wvel,uvel);
56
57 h      = -PD;
58 rho    = isa_density(h);     % ISA atmosphere density (see local function below)
59
60 qbar   = 0.5*rho*V^2;        % dynamic pressure
61 qhat   = q*cbar/(2*V);
62
63 % Altitude-varying gravity
64 g      = g0*(Re/(Re+h))^2;
65
66 %-----AERODYNAMIC COEFFICIENTS, ALPHA-DOT-INDEPENDENT PART
67
68 -----
69 % Split CL, Cm into a part independent of alpha_dot ("base") and the
70 % coefficient multiplying alpha_dot (adotthat = alpha_dot*cbar/(2V)).
71 CLbase = CL0 + CLa*alpha + CLq*qhat + CLde*de;
72 Cmbase = Cm0 + Cma*alpha + Cmq*qhat + Cmde*de;
73 CD      = CD0 + CDa*alpha + CDde*de;
74
75 Lbase  = qbar*S*CLbase;
76 D      = qbar*S*CD;
77 Lcoef  = qbar*S*CLadot*cbar/(2*V);           % dL/d(alpha_dot)
78 Mcoef  = qbar*S*cbar*Cmadot*cbar/(2*V);       % dMy/d(alpha_dot)
79
80 Mybase = qbar*S*cbar*Cmbase;
81
82 %-----PROPULSION FORCE
83
84 -----
85 SP = uth*Pmax;           % shaft power
86 T  = etap*SP/V;
87
88 % ROTATION MATRICES
89 Cwb = [ cos(alpha), 0, -sin(alpha);
90          0, 1, 0;
91          sin(alpha), 0, cos(alpha)];
92
93 % Aerodynamic force in wind axes, converted to body axes
94 Fwind_base = [-D; 0; -Lbase];
95 Fbody_base = Cwb*Fwind_base;

```

```

87
88 % d(F_aero_body)/d(alpha_dot)
89 dF_body_dadot = C_wb*[0;0;-Lcoef];
90
91 % Gravity rotation
92 Cbv = [ cos(theta), 0, -sin(theta);
93         0, 1, 0;
94         sin(theta), 0, cos(theta)];
95
96 Fg_v = [0; 0; m*g];
97 Fg_b = Cbv*Fg_v;
98
99 % -----TOTAL FORCE in Body Axes, No alpha-dot term
100 F_thrust_b = [T; 0; 0];
101
102 F_total_base = F_body_base + F_thrust_b + Fg_b; % alpha-dot not included here
103 Fx_base = F_total_base(1);
104 Fz_base = F_total_base(3);
105 Fx_coef = dF_body_dadot(1);
106 Fz_coef = dF_body_dadot(3);
107
108 % -----TOTAL MOMENT in Body Axes, No alpha-dot term
109 My_coef = Mcoef;
110
111 %
112 % -----
113 % 3x3 linear system solving the coupled alpha-dot implicit relationship
114 %
115 % udot = Fx_base/m - q*w_vel + (Fx_coef/m)*alpha_dot ... (1)
116 % wdot = Fz_base/m + q*u_vel + (Fz_coef/m)*alpha_dot ... (2)
117 % alpha_dot*V^2 = u_vel*wdot - w_vel*udot ... (3)
118 %
119 % -----
120
121 A1 = Fx_base/m - q*w_vel;
122 B1 = Fx_coef/m;
123 A2 = Fz_base/m + q*u_vel;
124 B2 = Fz_coef/m;
125
126 Mmat = [1, 0, -B1;
127         0, 1, -B2;
128         -w_vel, u_vel, -V^2];
129 rhs = [A1; A2; 0];
130
131 sol = Mmat\rhs;
132 udot = sol(1);
133 wdot = sol(2);
134 alpha_dot = sol(3);
135
136 % Final My including the alpha_dot contribution
137 My = My_base + My_coef*alpha_dot;
138
139 % Rotational (simplified - only Iyy relevant for longitudinal 3DOF)
140 qdot = My/Iyy;
141
142 % Angular position (simplified H(Phi) for phi=0)
143 thetadot = q;
144
145 % Translational position (simplified Cvb for phi=0,psi=0)
146 PNdot = u_vel*cos(theta) + w_vel*sin(theta);

```

```

144 PDdot = -u_vel*sin(theta) + w_vel*cos(theta);
145
146 % OUTPUT
147 xdot = [udot; wdot; qdot; thetadot; PNdot; PDdot];
148
149 end
150
151 %=====
152 function rho = isa_density(h)
153 % Simple ISA troposphere model (valid up to 11000 m)
154 rho0 = 1.225; % kg/m^3, sea level
155 T0 = 288.15; % K, sea level
156 L = 0.0065; % K/m, temperature lapse rate
157 R = 287.05; % J/(kg*K)
158 g0 = 9.80665; % m/s^2
159
160 T = T0 - L*h;
161 rho = rho0*(T/T0)^(g0/(R*L) - 1);
162 end

```

8 Simple Motion Visualization

As a first sanity check of the model (prior to trim analysis), the aircraft was simulated open-loop from an arbitrary, non-trimmed initial condition ($u_0 = 85 \text{ m/s}$, $w_0 = 0$, $\theta_0 = 3^\circ$, $h_0 = 1,524 \text{ m}$) with a fixed, non-equilibrium control input ($\delta_e = -2^\circ$, $u_{th} = 0.6$). Figure 2 shows the resulting flight path in the North-position/altitude plane, exhibiting the expected transient climb-and-curve behavior of an aircraft released away from its trim condition – confirming that the equations of motion integrate to a smooth, bounded, physically plausible trajectory before any trim or control-design work is attempted.

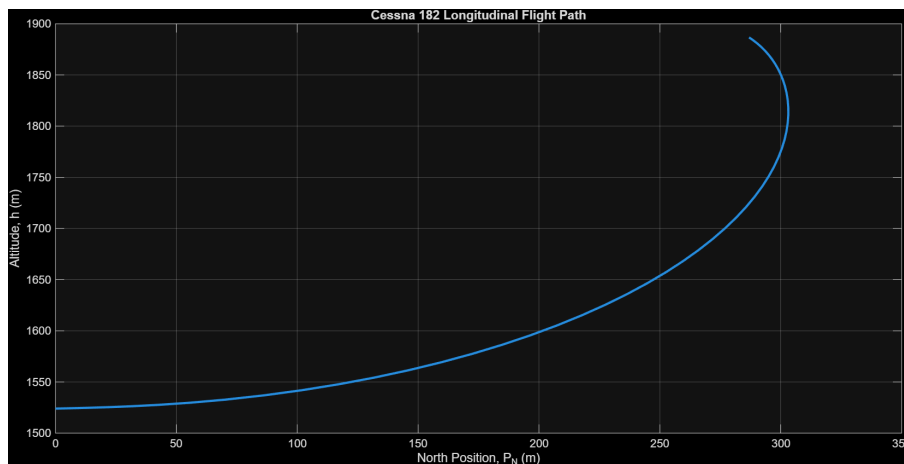


Figure 2: Sample open-loop flight path from a non-trimmed initial condition, illustrating basic model motion.

9 Trim Analysis

9.1 Method

Trim (equilibrium) conditions for steady, straight, and level flight are found by posing the problem as a numerical optimization. Let $\bar{z} = [u, w, q, \theta, \delta_e, u_{th}]^T$ be the vector of unknowns

for a given altitude h and target true airspeed V_{target} . The steady, straight, and level flight condition is equivalent to the equality constraints

$$\dot{u} = 0, \quad \dot{w} = 0, \quad \dot{q} = 0, \quad V - V_{\text{target}} = 0, \quad \gamma = \theta - \alpha = 0, \quad q = 0. \quad (29)$$

Rather than an equality-constrained solver, these constraints are enforced via a quadratic penalty (soft-constraint) approach: an unconstrained scalar cost function

$$f_0(\bar{z}) = w_{\dot{u}}\dot{u}^2 + w_{\dot{w}}\dot{w}^2 + w_{\dot{q}}\dot{q}^2 + w_z(\text{alt. rate})^2 + w_{\text{speed}}(V - V_{\text{target}})^2 + w_{\theta}\theta^2 \quad (30)$$

is minimized directly with `fminsearch` (derivative-free Nelder–Mead simplex), with all penalty weights set to unity. For each target airspeed, the trim solution from the previous (adjacent) airspeed is used as the initial guess, which improves convergence speed and robustness across the sweep.

Listing 2: `trim_cessna.m` – trim sweep over TAS = 30–80 m/s at 5,000 ft.

```

1 clear
2 clc
3 close all
4
5 h = 5000*0.3048; % 5000 ft to SI units
6 TAS_list = [30 40 50 60 70 80];
7
8 % First guess: [u; w; q; theta; de; uth]
9 Z_guess = [TAS_list(1); 0; 0; 0; 0; 0];
10
11 results = zeros(6,9);
12
13 for i = 1:length(TAS_list)
14     Va_target = TAS_list(i);
15
16     costfun = @(Z) cost_trim_cessna(Z, Va_target, h);
17
18     [ZStar,f0] = fminsearch(costfun, Z_guess, ...
19         optimset('TolX',1e-10,'TolFun',1e-12,'MaxFunEvals',20000,'MaxIter',
20             20000));
21
22     u_vel = ZStar(1);
23     w_vel = ZStar(2);
24     q = ZStar(3);
25     theta = ZStar(4);
26     de = ZStar(5);
27     uth = ZStar(6);
28
29     XStar = [u_vel; w_vel; q; theta; 0; -h];
30     UStar = [de; uth];
31
32     % Validation: state derivative should be (numerically) zero at trim
33     XdotStar = cessna182_model(XStar,UStar)
34
35     Va = sqrt(u_vel^2 + w_vel^2);
36     alpha = atan2(w_vel,u_vel);
37     gam = theta - alpha;
38
39     % Dynamic pressure and aerodynamic coefficients at the trim point
40     rho = isa_density(h);
41     qbar = 0.5*rho*Va^2;
42
43     cbar = 1.49;
44     qhat = q*cbar/(2*Va);

```

```

45 CL = 0.307 + 4.41*alpha + 3.9*qhat + 0.43*de;
46 CD = 0.0270 + 0.121*alpha;
47 Cm = 0.04 - 0.613*alpha - 12.4*qhat - 1.122*de;
48
49
50 results(i,:) = [Va, rad2deg(alpha), rad2deg(theta), rad2deg(gam), ...
51                qbar, CL, CD, Cm, rad2deg(de)];
52
53 fprintf('TAS=%d m/s -> alpha=%.3f deg theta=%.3f deg de=%.3f deg uth
54         =%.4f f0=%.3e\n', ...
55         TAS_list(i), rad2deg(alpha), rad2deg(theta), rad2deg(de), uth, f0);
56
57 % Use this solution as the initial guess for the next airspeed
58 Z_guess = ZStar;
59
60 save(['trim_' num2str(TAS_list(i)) 'ms.mat'], 'XStar', 'UStar');
61 end
62
63 T = array2table(results, 'VariableNames', ...
64   {'TAS', 'alpha_deg', 'theta_deg', 'gamma_deg', 'qbar', 'CL', 'CD', 'Cm', 'de_deg'})
65   ;
66 disp(T)

```

Convergence was verified at every airspeed by re-evaluating $\dot{x}$ at the converged solution using the full nonlinear model (`cessna182_model.m`) and confirming that all six state derivatives are numerically zero (residuals on the order of 10^{-11} or smaller; see terminal output accompanying Table 2), and that the optimizer's cost function value f_0 converged to below 10^{-20} at every airspeed.

9.2 Results

Table 2: Trim (equilibrium) values for straight-and-level flight at 5,000 ft, TAS = 30–80 m/s.

TAS [m/s]	α [°]	θ [°]	γ [°]	$\bar{q}$ [Pa]	C_L	C_D	C_m	δ_e [°]
30	16.341	16.341	≈ 0	475.0	1.5131	0.0615	≈ 0	-6.885
40	7.321	7.321	≈ 0	844.4	0.8558	0.0425	≈ 0	-1.957
50	3.116	3.116	≈ 0	1319.4	0.5494	0.0336	≈ 0	0.340
60	0.824	0.824	≈ 0	1900.0	0.3824	0.0287	≈ 0	1.592
70	-0.561	-0.561	≈ 0	2586.1	0.2815	0.0258	≈ 0	2.349
80	-1.460	-1.460	≈ 0	3377.7	0.2159	0.0239	≈ 0	2.840

At every trim point, $\gamma = \theta - \alpha \approx 0$ (to within 10^{-9} degrees), confirming true straight-and-level flight, and $C_m \approx 0$ (to within 10^{-12}), confirming moment balance about the pitch axis. The monotonic decrease in α and increase in δ_e and u_{th} with increasing airspeed are consistent with the expected physics: less lift coefficient is required as dynamic pressure increases, requiring progressively less nose-up trim, and progressively more thrust to balance the growing drag.

These results were independently cross-checked by solving the same two governing equations ($C_m = 0$ and vertical force balance) directly with a separate root-finding routine outside of the optimization-based pipeline; the two approaches agree to within the solver tolerances, confirming the trim table's internal consistency and correctness.

10 Elevator Doublet Simulation

10.1 Simulink Model

The nonlinear model is embedded in a Simulink block diagram (Fig. 3): `cessna182_model.m` is wrapped as an Interpreted MATLAB Function block, integrated with a $1/s$ block, with the resulting state fed back to close the loop (via a Mux, forming the combined $[x; u]$ input required by the function block), and logged to the workspace as `simX/simU`.

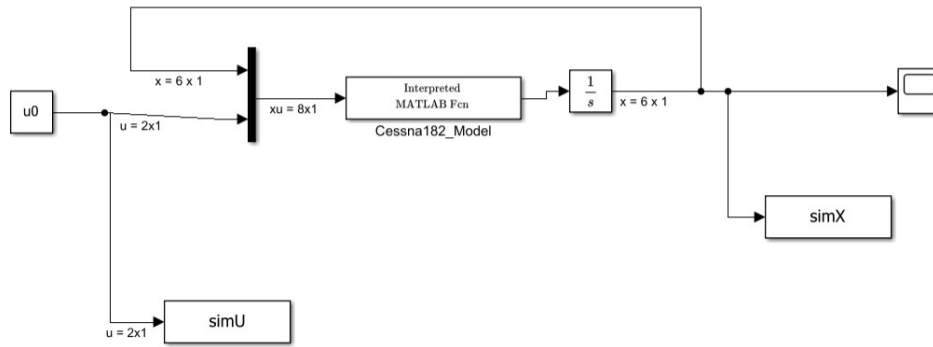


Figure 3: Base Simulink model: state/control multiplexing, nonlinear model evaluation, integration, and logging.

Starting from the 60 m/s trim condition of Table 2 ($\bar{X}^* = [u^*, w^*, q^*, \theta^*, 0, -h]^T$, $\bar{U}^* = [\delta_e^*, u_{th}^*]^T$, saved to `trim_60ms.mat`), a perturbation subsystem (Fig. 4) superimposes an elevator doublet on top of the trim control vector $\bar{U}^*$: a $+5^\circ$ step at $t = 1$ s, a -10° step at $t = 5$ s (giving a net -5° level), and a $+5^\circ$ step at $t = 9$ s returning the command to the trim value, matching the reference doublet shape specified for this study. Because a Step block produces an instantaneous transition, a Rate Limiter is placed on the summed elevator signal to reproduce the finite-slew-rate, ramped transitions of the reference doublet profile (physically consistent with a real elevator servo's bounded angular rate) rather than a discontinuous jump.

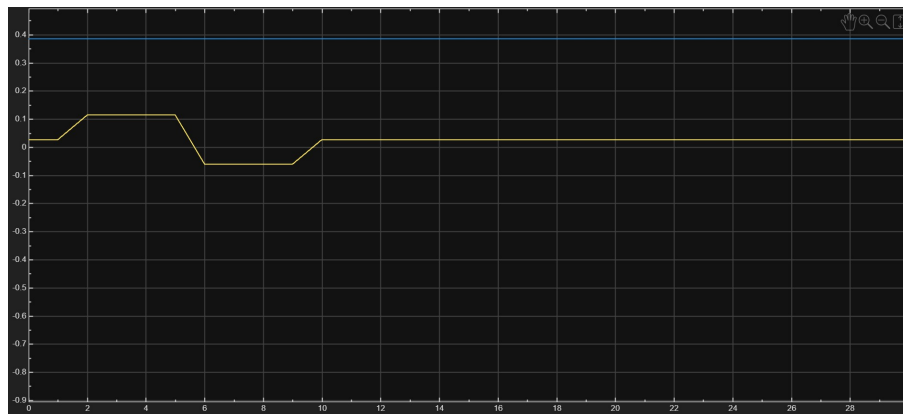


Figure 4: Applied control inputs over the 30 s simulation: elevator command δ_e (yellow), rate-limited about the trim value, and throttle u_{th} (blue, held constant at its trim value throughout the maneuver).

10.2 Solver Configuration

The variable-step `ode45` solver is used with a reduced maximum step size (rather than `auto`) to ensure the rate-limited ramps are adequately resolved; with the default automatic step-size selection, the solver under-samples the short (~ 1 s) ramp transitions, producing visible “stair-casing” in the logged elevator signal. Explicitly bounding the maximum step size (on the order of 10^{-3} s) resolves this while remaining computationally inexpensive, since the underlying aircraft dynamics themselves are comparatively slow.

Table 3: Key ODE45 solver settings used for the doublet simulation.

Setting	Value
Solver	<code>ode45</code> (variable-step Runge–Kutta 4/5)
Maximum step size	$\sim 10^{-3}$ s (explicitly bounded, not <code>auto</code>)
Simulation duration	30 s, from the 60 m/s trim solution

10.3 Results

Figure 5 shows the six state time histories over the 30-second simulation. The response exhibits the classical signature of a lightly-damped phugoid mode: u and altitude ($-P_D$) oscillate out of phase (kinetic/potential energy exchange), with the short-period response visible as the rapid initial transient in q coincident with the doublet input itself ($t = 1$ – 10 s).

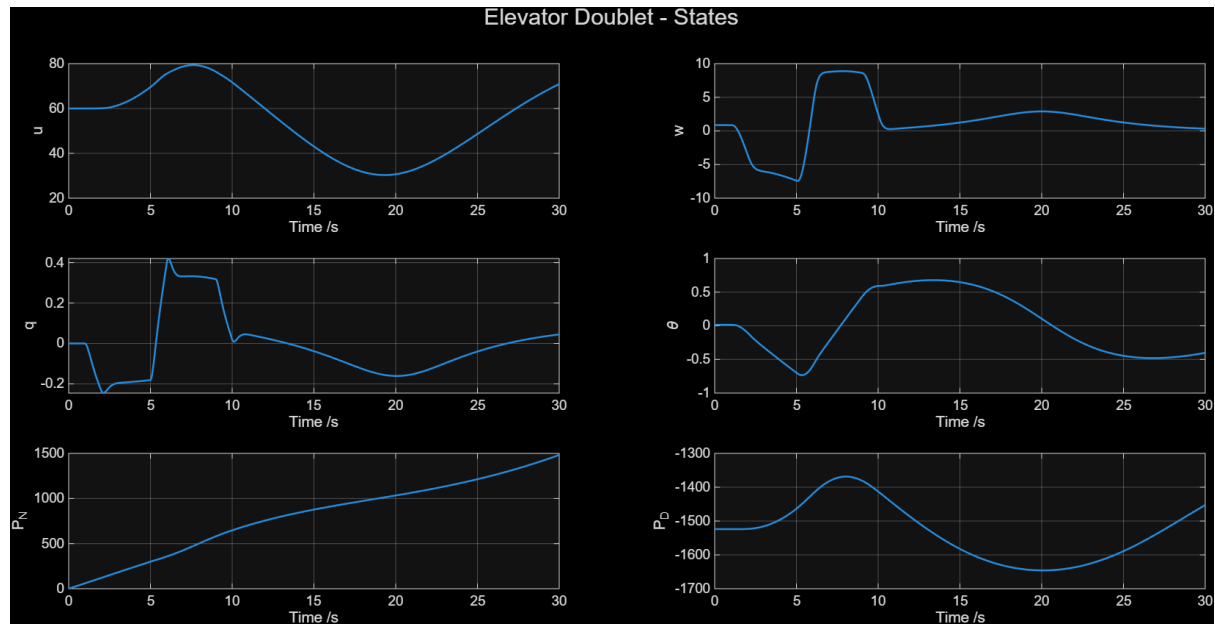


Figure 5: State time histories following a $\pm 5^\circ$ elevator doublet applied from the 60 m/s trim condition.

A longer (100 s) simulation of the same maneuver (not shown for the required 30 s window, but used as an internal verification step) confirms that the oscillation amplitude decreases monotonically over successive cycles and the aircraft returns toward its original trim condition, consistent with the negative static longitudinal stability derivative $C_{m_\alpha} = -0.613 < 0$ and providing strong independent verification of the model’s dynamic correctness beyond the trim-point checks of Section 9.

11 Conclusion

A nonlinear, 3-DOF longitudinal flight dynamics model of the Cessna 182 was developed from first principles, starting from the general flat-Earth rigid-body equations of motion and systematically reducing them under an explicit, documented set of physical assumptions. The model's principal technical contribution is the closed-form resolution of the coupled $\dot{\alpha}$ algebraic loop, which is solved exactly as a 3×3 linear system rather than approximated – avoiding the reproducibility and correctness issues associated with delayed-estimate or neglected-term approaches. The model was verified at six independent trim points (30–80 m/s at 5,000 ft, straight-and-level flight, residuals below 10^{-10}) and further verified dynamically via a $\pm 5^\circ$ elevator doublet maneuver, which produced the expected lightly-damped phugoid response and a return toward the original trim state – both hallmarks of a correctly implemented, statically stable aircraft model.

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